

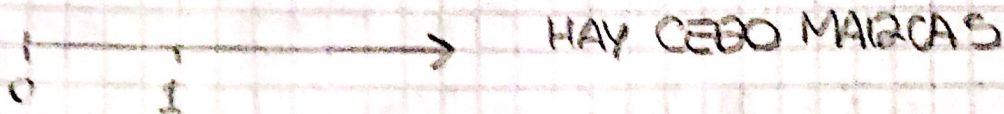
GUÍA - 7

7.1

$\{N(t), t \geq 0\}$ PROCESO DE CONTEO $\sim \text{POI}(2)$

$N(A, B) = N(B) - N(A)$ INCREMENTO DEL PROCESO.

A-



$$N(t) \sim \text{POI}(2t) \quad \mu = 2 \quad t = 1$$

$$P(N(1) = 0)$$

$$P_x(0) = \frac{2^0 e^{-2}}{0!} = e^{-2}$$

$$B - P(N(1, 2) = 1)$$

$$N(1, 2) = N(1)$$

$$P(N(1) = 1) \quad N(1) \sim \text{POI}(2)$$

$$P_x(1) = \frac{2^1 e^{-2}}{1!} = 2e^{-2}$$

$$C - P(N(1) = 0, N(1, 2) = 1, N(2, 4) = 2)$$

$$P(N(1) = 0, N(1) = 1, N(2) = 2)$$

$$P_x(2) = \frac{(2)^2 e^{-2}}{2!} = 2e^{-2} \quad N(2) \sim P(4)$$

$$P(N(1)=0, N(1)=1, N(2)=2) = \text{COMO SON INDEP}$$

$$P(N(1)=0) P(N(1)=1) P(N(2)=2)$$

$$e^{-2} \cdot 2e^{-2} \cdot 8e^{-4} = 16e^{-8}$$

$$D- \text{COV}(N(1,3), N(2,4)) =$$

$$\text{COV}[(N(1,2) + N(2,3)), (N(2,3) + N(3,4))]$$

$$\text{COV}(N(1,2), N(2,4)) + \text{COV}(N(1,2), N(3,4))$$

$$\text{COV}(N(2,3), N(2,3)) = \text{VA}(N(2,3))$$

$$\text{VA}(N(1)) = \lambda \cdot T = 2$$

S_3 TIEMPO DE ESPERA HASTA EXITO 3:

E-



LA SUMA ES
 $P(S_3 > 1/2)$

$$\textcircled{1} N(1/2) = 2 + \textcircled{2} N(1/2) = 1 + N(1/2) = 0$$

$$e^{-1} + e^{-1} + \frac{1}{2} e^{-1} = \frac{5}{2} e^{-1}$$

7.2

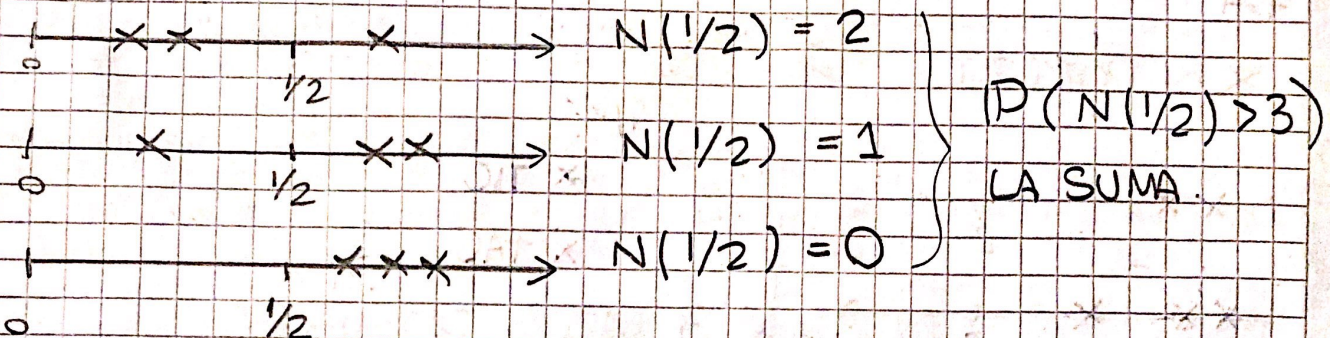
 $X = \# \text{ ACCIDENTES }$

RECORDAR QUE

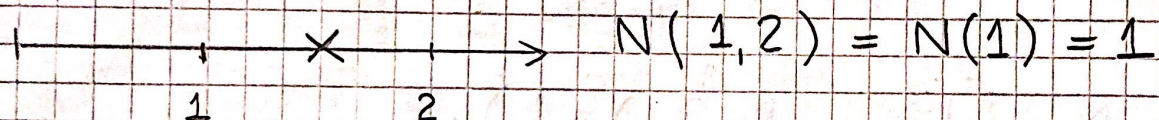
 $\sim \text{POI} (2T)$

$$X \sim \text{POI} (1) \quad \lambda = \frac{2 \text{ ACCID}}{1 \text{ HS}}$$

$$A - P(X > 1/2)$$



B-



$$P(N(1) = 1) = \frac{(1.2)^1 e^{-(1.2)}}{1!} = 2e^{-2}$$

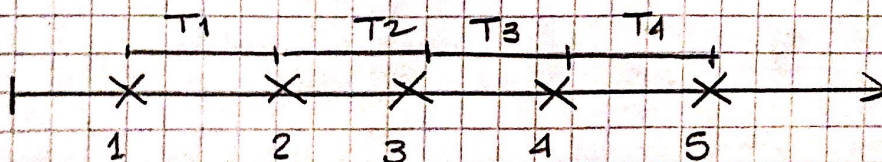
7.3

 $X(t): \# \text{ PARTICULAS EMITIDAS }$

$$X \sim \text{POI} (10T) \quad \lambda = \frac{10 \text{ PART}}{1 \text{ HS}}$$

EMISOR

$$2 \text{ MÍN} = \frac{1}{30}$$

 $T: t \text{ ENTRE MARCAS} \sim \text{EXP} \left(\frac{10 \text{ PART}}{\text{HS}} \right)$


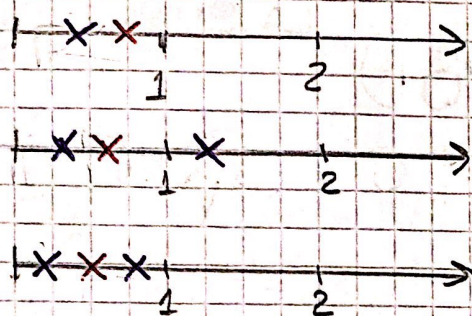
$$P\left(T_1 > \frac{1}{30}, T_2 > \frac{1}{30}, T_3 > \frac{1}{30}, T_4 > \frac{1}{30}\right)$$

COMO SON INTERVALOS $\neq \rightarrow$ SON DISJUNTOS

$$\begin{aligned} 5. \quad P\left(T > \frac{1}{30}\right) &= 5 \cdot \left(1 - P\left(T \leq \frac{1}{30}\right)\right) \\ &= 5 e^{-10 \cdot \frac{1}{30}} = 0,2636 \end{aligned}$$

7.4

$$X \sim \text{Poi}^2(2T) \quad \lambda = 2 \frac{\text{PART}}{1 \text{ SEG}}$$



x TIC

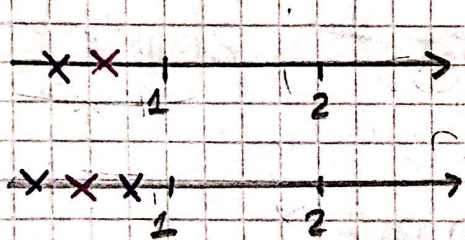
x TAC

3 CASOS POSIBLES

$$P(N(1,2)=0 \mid N(0,1)=1)$$

$$\begin{aligned} &P(N(0,1)=2, N(2,1)=0) + P(N(0,1)=2, N(1,2)=1) \\ &+ P(N(0,1)=3, N(1,2)=0) \end{aligned}$$

SOLO LOS CASOS EN LOS QUE $N(0,1)=1$



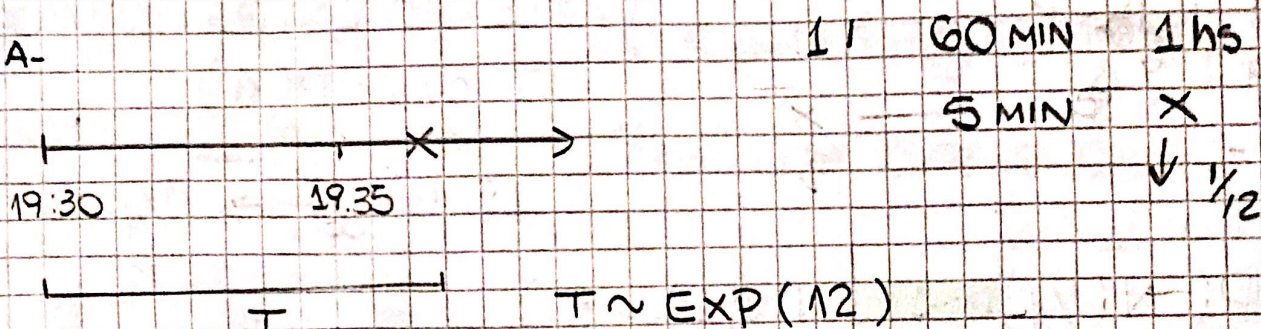
$$P(N(0,1)=2) + P(N(0,1)=3)$$

$$= 0,2077$$

7.5

COLECTIVO LLEGA $\sim \text{POI}(\frac{12}{\text{hs}})$

A-



$$\begin{aligned} P(T > \frac{1}{12}) &= 1 - P(T \leq \frac{1}{12}) \\ &= 1 - (1 - e^{-12 \cdot \frac{1}{12}}) \\ &= e^{-1} \end{aligned}$$

B- ES LO MISMO QUE (A)

C- SIGUE SIENDO LO MISMO PORQUE NO INTERESA EL T INICIAL SE CUENTA S DESDE AHÍ.

D-

LLEGADA $19:00 + U$ $U \sim U(0, 60)$

$$f_U(u) = \frac{1}{60} \mathbb{I}_{\{0 \leq u < 60\}}$$

7.6

EL PESO DE BOLSAS $= X$ $X \sim \text{EXP}(\frac{1}{3})$
SE AGREGA HASTA 5 kg .

$$P(S_N > 7 \mid S_N > 5) = P(5 + Y > 7) = P(Y > 2)$$

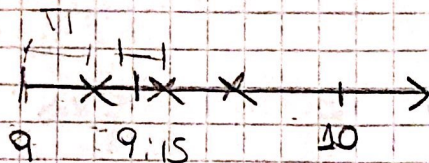
$$P(Y > 2) = 1 - P(Y \leq 2) = e^{-2/3}$$

B- TODAS LAS BOLSAS TIENEN LA MISMA MEDIA $= 3$

7.7

$X = \text{'# LLAMADAS QUE ARRIBAN'}$

$$X \sim \text{Poi}(4) \quad \lambda = \frac{4 \text{ LLAMADAS}}{1 \text{ hs}}$$



$T = \text{'TIEMPO ENTRE LLAMADAS'}$
 $T \sim \text{EXP}(4)$

A -

$$15 \text{ MIN} \longrightarrow \frac{1}{4} \text{ hs}$$

$$P(N(\frac{1}{4}) \geq 1 \mid N(0,1) = 3)$$

$$P(N(\frac{1}{4}) \geq 1 \mid N(1) = 3) =$$

$$\frac{P(N(\frac{1}{4}) \geq 1) \cdot P(N(\frac{3}{4}) \leq 2)}{P(N(1) = 3)}$$

SEPARO
EL INTERVALO
PARA QUE SEAN
INDEP

$$X = (N(\frac{1}{4}) \mid N(1) = 3)$$

$$X \sim \text{BIN}(3, \frac{1/4}{1})$$

PROPIEDAD!!

X ES UNIFORME

$$P(X \geq 1) = 1 - P(X < 1) = 1 - P_X(0)$$

$$= 1 - \binom{3}{0} \left(\frac{1}{4}\right)^0 (1 - \frac{1}{4})^3 = 0,58$$

B - $P(N(0, \frac{1}{2}) \geq 2 \mid N(0,1) = 3)$

$$\frac{P(N(\frac{1}{2}) \geq 2) \cdot P(N(1) \leq 1)}{P(N(1) = 3)}$$

$$N(\frac{1}{2}) \mid N(1) = 3 = Y \quad Y \sim \text{BIN}(3, \frac{1}{2})$$

NOTA

$$P(Y \geq 2) = 1 - P(Y < 2) = \frac{1}{2}$$

7.8

X : "# DE PERSONAS QUE LLEGAN A LA ESTACIÓN"

$$X \sim PO^? (50) \quad \lambda = \frac{50 \text{ P}}{1 \text{ MIN}}$$

• T = "# TOTAL DE TIEMPO QUE ESPERARON LOS P"

• $T = \sum_{i=1}^{N(0,15)} T_i$ SIENDO T_i LO QUE ESPERO CADA PASAJERO.

$$T = \sum_{i=1}^{N(15)} T_i \quad \text{PROCESO DE POSON COMPUESTO}$$

S_i = INSTANTE DE ARRIBO DE CADA PASAJERO

$$T = \sum_{i=1}^{N(15)} (15 - S_i) \quad T_i = 15 - S_i$$

$$(T_i = 15 - S_i \mid N(0,15) = n) \sim U(0,15)$$

$$E[T] = E[E[T \mid N(0,15)]]$$

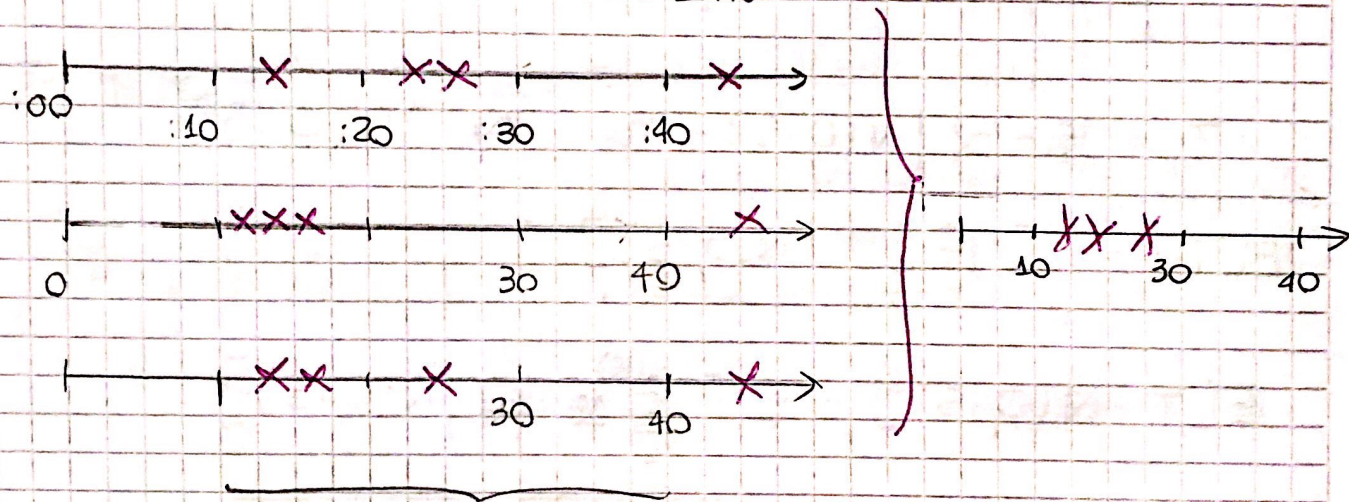
$$E[T \mid N(0,15)] = E \left[\sum_{i=1}^{N(15)} 15 - S_i \mid N(0,15) = n \right]$$

$U(0,15)$

7.9

$X = \text{'\# EMISION DE PARTICULAS'}$

$$X \sim \text{poi}(\lambda) \text{ con } \lambda = \frac{6p}{1\text{ns}}$$



EN VERDAD YO SE QUE DE 10-40 HAY 3

$$P(N(0,10)=0 \cap N(30,40)=0 \mid N(0,30)=3)$$

$$P(N(0,10)=0 \cap N(10,30)=3 \cap N(30,40)=0 \mid N(0,30)=3)$$

$$P(N(0,10)=0 \cap N(30,40)=0 \mid N(0,30)=3)$$



$$P(N(0,10)=0 \mid N(0,30)=3) \cap (N(30,40)=0 \mid N(0,30)=3)$$

NO ES INDEP, ENTONCES
LO HAGO

YA ES INDEP
ENTONCES

$$\frac{P(N(0,10)=0) \cdot P(N(10,30)=3)}{P(N(0,30)=3)} \cdot P(N(30,40)=0)$$

X =

$$Z \sim \text{BIN}(3, \frac{10}{30})$$

(10-0) INTERVALO EXITO

TOTAL

$$P(Z=0) \cdot P(N(10)=0) \rightarrow \frac{6}{11s}$$

$\sim \text{Poi}(10 \cdot \frac{1}{10})$

$$P(Z=3) \cdot P(N(10)=0) \quad Z \sim \text{BIN}(3, \frac{20}{30})$$

$$P(\text{FINAL}) = \frac{8}{27} \cdot e^{-1} = 0,109$$

7.10

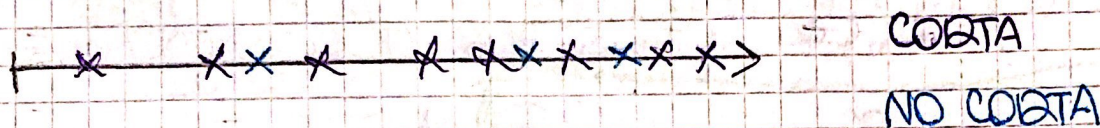
$X =$ FALLAS DEL ALAMBRE

$$X \sim \text{Poi}\left(\frac{1F}{20m}\right)$$

A-

MAQUINA DETECTA $\rightarrow$ UNA FALLA $\rightarrow P_F = 0,75$

$\rightarrow$ CORTA EL ALAMBRE EN LA 1 FALLA



$$\pi_1 = \{ \text{CORTA} \} \sim \text{Poi}\left(0,75 \cdot \frac{1}{20}\right)$$

$$\pi_2 = \{ \text{NO CORTA} \} \sim \text{Poi}\left(0,25 \cdot \frac{1}{20}\right)$$

LONGITUD $\rightarrow$ METROS HASTA EL PRIMER CORTE

$$\text{LONGITUD} \sim \text{EXP}\left(0,75 \cdot \frac{1}{20}\right)$$

$$E[L] = \frac{1}{0,75} \cdot 20 = \frac{80}{3}$$

$$\text{VAR}(L) = \frac{1}{0,75^2} = \frac{6400}{9}$$

B-

$$E[\text{FALLAS EN UN ROLLO}] = E[\pi_2]$$